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Next item →

1. What is the meaning of $\hat{f}(\mathbb{X}_k \mid \mathbb{Z}_k)$?

1 / 1 point

- ☐ It is an estimate of the pdf of the most recent state vector x_k conditioned on all measurements up to and including those at time k .
- ☒ It is an estimate of the pdf of the entire state trajectory $\{x_0, x_1, \dots, x_k\}$ conditioned on all measurements up to and including those at time k .
- ☐ It is an estimate of the value of most recent state vector x_k conditioned on all measurements up to and including those at time k .
- ☐ It is an estimate of the value of entire state trajectory $\{x_0, x_1, \dots, x_k\}$ conditioned on all measurements up to and including those at time k .

✔ **Correct**

Yes, this is correct.

2. What is the reasoning behind writing the importance density as:

1 / 1 point

$$q(\mathbb{X}_k \mid \mathbb{Z}_k) = q(x_k \mid x_{k-1}, z_k)q(\mathbb{X}_{k-1}, \mathbb{Z}_{k-1})?$$

- ☐ This is the result of a mathematical derivation. All conditional pdfs factor into this form.
- ☐ We choose a pdf that factors this way because otherwise we cannot draw random samples from the distribution.
- ☒ This is a design choice made when selecting the importance density. It makes the particle-filter algorithm recursive.
- ☐ We choose a pdf that factors this way because otherwise the Monte-Carlo method does not give the correct answer.

✔ **Correct**

Yes, this is true.

3. If we desire to track the marginal pdf (instead of the joint pdf), what are the particle-filter steps?

1 / 1 point

- ☐ Draw an entire new set of random trajectories; update the weights for each trajectory.
- ☐ Draw an entire new set of random trajectories; keep only the final element of each trajectory; update the weights for each trajectory.
- ☒ Draw one new random state for each trajectory; update the weights for each trajectory.
- ☐ Draw an entire new set of random trajectories; integrate the joint pdf to find the marginal pdf for each trajectory; update the weights for each trajectory.

✔ **Correct**

Yes, this is correct.